https://github.com/Xuanxuan-zi/Machine-learning-Yuxuan-Gou

Perceived Stress as a Social and Psychological Outcome: The Role of Socioeconomic Status, Psychosocial Resources, and Adverse Life Experiences

1. Introduction

Perceived stress is an important part of mental health and daily life. When people feel high levels of stress, they may have more emotional problems, worse physical health, and more difficulty in everyday functioning. Stress is not only a personal feeling. It is also related to people's social conditions, health status, family environment, and life experiences[1].

Many studies have examined factors related to stress. Some studies focus on sociodemographic factors such as age, gender, education, income, and employment[2]. Other studies show that psychosocial resources, such as self-efficacy, family support, and social support, may protect people from stress[3]. In addition, stressful life experiences, including major life events and adverse childhood experiences, are often associated with higher stress. Anxiety and depression are also closely related to perceived stress.

Although previous studies have identified many possible correlates of stress, fewer studies have compared these different groups of factors in one framework. In particular, it is still useful to ask whether perceived stress is more strongly shaped by basic demographic background or by more proximal factors such as psychological distress, psychosocial resources, and adverse life experiences.

This paper examines which sociodemographic, psychosocial, and life-course factors matter most for perceived stress. The outcome is pss4_total. To address this question,

I combine OLS regression, LASSO, and Random Forest, and then use SHAP values to improve the interpretation of the machine learning model. Looking across these approaches helps show which predictors remain important across different modeling strategies.

The main research question is:

1） Which sociodemographic, psychosocial, and life-course factors are most important in explaining and predicting perceived stress?

2） More specifically, this study asks whether psychological distress, psychosocial resources, and adverse life experiences contribute more strongly to perceived stress than basic demographic characteristics alone.

2. Literature Review

2.1 Perceived Stress as a Social and Psychological Outcome

Perceived stress refers to the extent to which individuals feel that their lives are unpredictable, uncontrollable, or overwhelming. It is a psychological outcome, but it is also shaped by social conditions[4]. People do not experience stress in the same way, because they live in different social and material environments. For this reason, perceived stress can be understood as both an individual psychological experience and a socially patterned outcome.

Previous research has shown that high perceived stress is associated with poorer mental health, lower well-being, and worse physical health. Stress is also related to how people understand and respond to challenges in daily life. Because of these broad consequences, perceived stress is an important outcome to study in both psychology and social science[5].

2.2 Sociodemographic Factors and Perceived Stress

Many studies have examined the relationship between sociodemographic characteristics and stress. Factors such as age, gender, education, income, marital status, and employment status are often used to explain differences in stress levels. In general, people with fewer socioeconomic resources may face more uncertainty and pressure in daily life, which can increase stress[6].

Income is especially important because it shapes access to material resources and economic security. People with lower income may face more financial strain and fewer coping resources[7]. Employment status can also matter, because work can provide income and social roles, but unemployment or unstable work may increase stress. Marital status and living arrangement may also influence stress by changing the level of social support and daily responsibilities available to individuals[8].

However, the effects of demographic characteristics are not always strong or consistent across studies. This suggests that demographic background alone may not be enough to explain perceived stress. More proximal psychological and social factors may play a more direct role.

2.3 Psychosocial Resources as Protective Factors

Psychosocial resources are often seen as protective factors against stress. These resources include self-efficacy, family support, social support, and resilience. They may help individuals cope with difficulties and reduce the negative effects of stressful situations[9].

Self-efficacy refers to a person's belief in their own ability to deal with problems and challenges. People with higher self-efficacy may feel more control over their lives, which can reduce perceived stress. Family support and social support are also important, because supportive relationships can provide emotional help, practical help,

and a sense of belonging. These resources may buffer the effects of stress and help people recover from difficult experiences[10].

Resilience is another resource that may reduce stress. It reflects the ability to adapt to hardship and bounce back from challenges. Although these psychosocial resources are conceptually different, they all suggest that stress is not only shaped by risk factors, but also by the resources people have to manage those risks[11].

2.4 Adverse Experiences and Psychological Distress

Adverse experiences across the life course may increase perceived stress. Major life events, financial difficulties, and adverse childhood experiences can create long-term emotional and social burdens. People who have experienced more adversity may become more vulnerable to stress later in life.

Adverse childhood experiences are especially important because they may influence emotional regulation, coping styles, and mental health over time[12]. Similarly, recent major life events may directly increase stress because they create uncertainty, loss, or disruption in daily life[13]. Debt and other financial strain may also make people feel less secure and more overwhelmed[14].

Psychological distress is also closely related to perceived stress. Anxiety and depression are often strongly associated with stress because they affect how people interpret challenges and how they respond to negative events[15]. People with higher anxiety or depressive symptoms may be more likely to perceive daily life as stressful. For this reason, anxiety and depression are likely to be among the strongest correlates of perceived stress.

2.5 Research Gap

Existing studies have shown that stress is related to demographic background, psychosocial resources, adverse experiences, and psychological distress. However,

these factors are often studied separately. Fewer studies have compared their relative importance within the same analytical framework.

In addition, many studies rely on only one modeling approach. Traditional regression models are useful for interpretation, but machine learning models may provide additional insight into prediction and variable importance. Therefore, this study brings these factors together and compares OLS, LASSO, and Random Forest models, with SHAP used to improve the interpretability of the machine learning results.

2.6 Research Question

This study asks: Which sociodemographic, psychosocial, and life-course factors are most important in explaining and predicting perceived stress?

More specifically, it examines whether:

1) psychological distress is positively associated with perceived stress,

2) psychosocial resources are negatively associated with perceived stress, and

3) adverse life experiences are associated with higher perceived stress.

3. Methodology

3.1 Data and sample

The data and subjects of this study are derived from the 2023 Psychology and Behavior Investigation of Chinese Residents (PBICR), a large-sample, multi-center, longitudinal, nationwide cross-sectional study conducted from June 20, 2023, to August 31, 2023. The survey design is scientific, with standardized sampling methods, comprehensive content, and strict quality control. It covers various aspects of public psychological (e.g., anxiety, depression, stress, personality, self-efficacy) and behavioral (e.g., smoking, drinking, exercise, diet, sleep) health, widely used in public health, psychology, sociology, etc. The survey employed a multi-stage sampling

method, including samples from 23 provinces and 5 autonomous regions of China, with a gender ratio of 1:1 and age distribution closely aligned with China's 'population pyramid'[16,17]. The sample size is 30054.

3.2 Variables

The main outcome of this study is perceived stress, measured by pss4_total. Higher scores indicate higher levels of perceived stress.

The predictor variables are grouped into four categories: sociodemographic characteristics, health and behavior factors, psychosocial resources, and adversity/life-course factors. Table 1 presents the definitions, coding, and analytic roles of the main variables.

In the main analyses, scale-based variables were treated as continuous variables. Categorical variables were coded as factor variables. Ordered variables, such as education and income, were coded so that higher values represent higher levels of socioeconomic resources.

Table 1. Measures and Variable Definitions

Category	Variable	Description	Coding / Range	Role in analysis
Outcome	pss4_total	Perceived stress total score (PSS-4)	0–16; higher = more perceived stress	Dependent variable
Sociodemographic	age	Respondent age	Continuous	Predictor
	gender	Gender	Categorical	
	education	Educational attainment	Ordered/continuous; higher = more education	
	income_ord	Household or personal income level	Ordered categories; higher = higher income	

	hukou	Hukou status	Agricultural / non-agricultural
	marital	Marital status	Categorical
	employment	Employment status	Categorical
	live_alone	Lives alone	Yes / No
Health & behavior	bmi	Body mass index	Continuous
	EQ_VAS	Self-rated health (EQ visual analogue scale)	0–100; higher = better health
	sleep_quality	Sleep quality score	Scale score; specify direction in text
	MEQtotal_new	Chronotype / morningness-eveningness score	Higher = more morningness
	drinking	Current drinking status	Yes / No
	smoking	Current smoking status	Yes / No
	chronic_disease	Presence of chronic disease	Yes / No
Psychosocial resources	social_support_total	Total social support score	Continuous
	self_efficacy_total	Total self-efficacy score	Continuous
	resilience_total	Total resilience score	Continuous
	family_support_total	Total family support score	Continuous
Adversity / life-course	any_life_event	Experienced any major life event	Yes / No
	debt_any	Has any debt	Yes / No

ACE_total	Adverse childhood experiences total	Continuous; higher = more ACE exposure	

3.3 Analytic strategy

This study uses several models to examine predictors of perceived stress.

First, I use OLS regression to examine the direction and statistical significance of the relationships between the predictors and perceived stress. This model provides a simple and interpretable baseline.

Second, I use LASSO regression to select a smaller set of important predictors. LASSO shrinks less important coefficients toward zero and helps identify a more parsimonious model.

Third, I use Random Forest to model perceived stress in a more flexible way. I estimate two Random Forest models: one using the full set of predictors and one using only the predictors selected by LASSO. The Random Forest models are tuned by cross-validation and evaluated on a held-out test set. To improve interpretability, I also use SHAP values for the full Random Forest model.

3.4 Model evaluation

Model performance is evaluated using RMSE, R^2, and MAE.

RMSE measures the average size of prediction errors. Lower RMSE means better predictive performance. R^2 shows how much variation in perceived stress is explained by the model. Higher R^2 means better model fit. MAE measures the average absolute prediction error.

For the Random Forest models, I first split the data into a training set and a test set. I then use 5-fold cross-validation on the training set to tune the hyperparameters. The final model is evaluated on the test set.

3.5 OLS diagnostics and robustness checks

Before estimating the main OLS model, I examined the distribution of the outcome variable and the standard regression diagnostic plots. Descriptively, perceived stress is not normally distributed. Scores range from 0 to 16, and lower values are relatively concentrated. This suggests some non-normality and a possible floor effect. However, because perceived stress scale is a scale-based variable with multiple ordered values, and because the sample size is large, it is still reasonable to treat it as an approximately continuous outcome in the baseline OLS analysis.

The diagnostic plots suggest that the OLS assumptions are not perfectly satisfied. The Q-Q plot shows that the residuals do not fully follow the reference line, especially in the tails. The Residuals vs Fitted and Scale-Location plots also suggest some heteroskedasticity. At the same time, the Residuals vs Leverage plot does not show a very extreme leverage pattern that would strongly drive the model results.

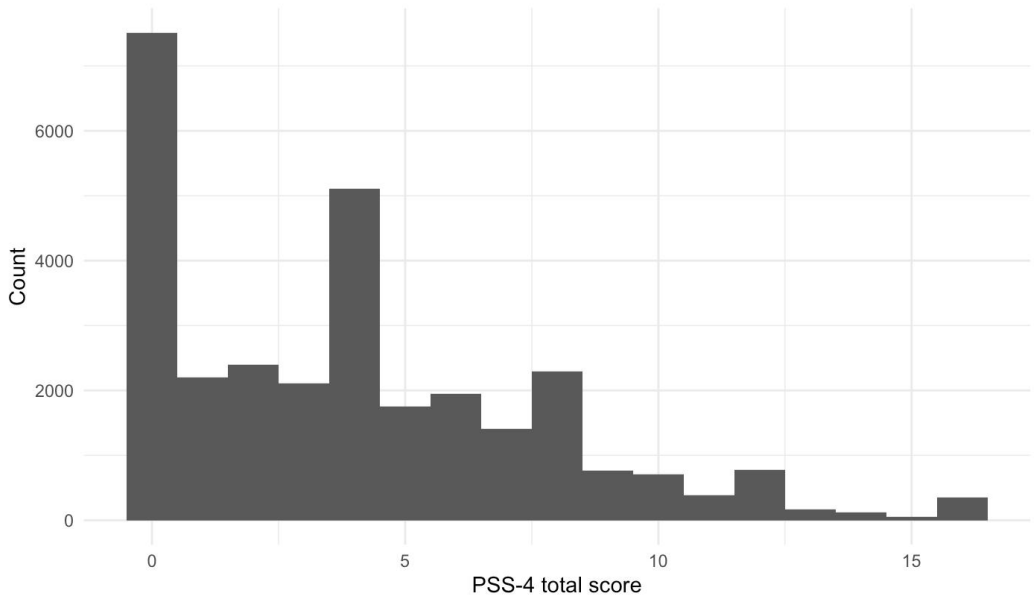

Figure 1: PSS-4 total score

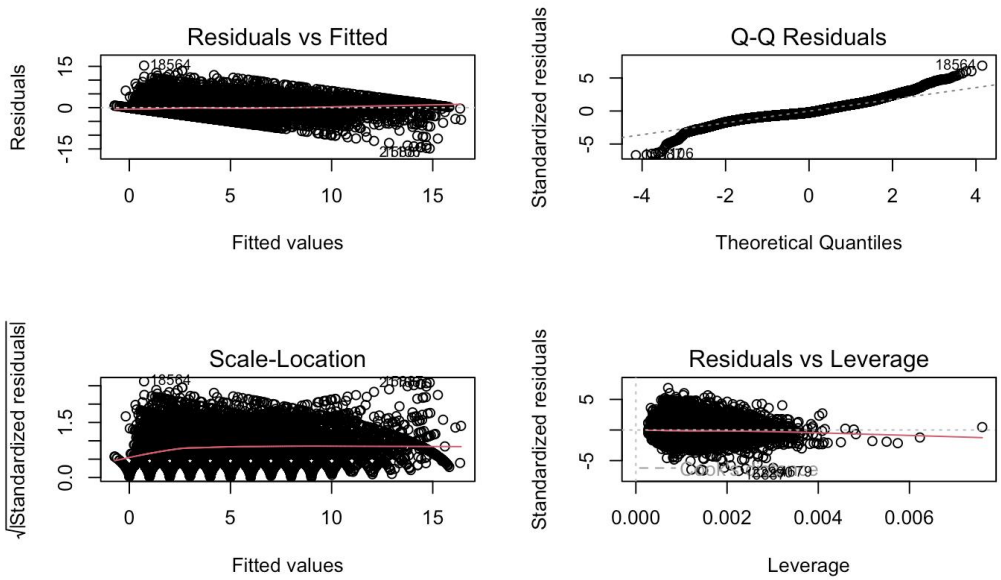

Figure 2: OLS Diagnostic Plots

To examine heteroskedasticity more formally, I conducted a studentized Breusch–Pagan test. Because heteroskedasticity was detected, I also examined HC3 robust standard errors. The main conclusions remained largely similar after this adjustment, suggesting that the OLS findings are reasonably robust. Therefore, OLS is retained as a baseline model, while the later LASSO and Random Forest models provide additional evidence from variable selection and prediction.

3.6 Interpretability

To improve the interpretation of the machine learning results, I use SHAP values for the full Random Forest model.

SHAP helps explain how each predictor contributes to the predicted level of perceived stress. It shows not only which variables are important, but also whether they tend to increase or decrease the predicted outcome. In this way, SHAP makes the Random Forest model easier to understand.

4.Result

4.1 Descriptive analysis

Table 2A and Table 2B summarize the main characteristics of the sample. Table 2A presents the frequency distributions of the main categorical variables, while Table 2B shows the distributions of the main continuous variables. Together, these tables provide an overview of the sample's sociodemographic background, health and behavioral conditions, psychosocial resources, and adverse life-course experiences. Table 2A shows that the sample is almost evenly split by gender and fairly balanced by hukou status. Most respondents are married, and employment is concentrated in the employed, flexible-job, student, and retired groups. In addition, the majority report that they do not live alone, have not experienced a major life event, and do not currently have debt, smoke, or drink.

Table 2A: Frequency Distribution of Main Categorical Variables

Variable	Category	N	Percent
Gender	female	15043	50.05
	male	15011	49.95
Hukou status	agricultural	15425	51.32
	non-agricultural	14629	48.68
Marital status	Divorced	675	2.25
	Married	19427	64.64
	Unmarried	8900	29.61
	Widowed	1052	3.5
Employment status	employed	13302	44.26
	student	4681	15.58
	unemployed	2800	9.32
	flexible_job	5043	16.78
	retired	4228	14.07
Lives alone	No	24605	81.87
	Yes	5449	18.13
Major life event	No	18536	61.68
	Yes	11518	38.32

Debt	No	18093	60.2
	Yes	11961	39.8
Drinking	No	22597	75.19
	Yes	7457	24.81
Smoking	No	25116	83.57
	Yes	4938	16.43
Chronic disease	No	20352	67.72
	Yes	9702	32.28

Table 2B: Distribution of Main Continuous Variables

Variable	Mean	SD	Min	Max
Perceived stress	4	3.68	0	16
Age	45.33	16.54	20	108
Education	4.95	2.07	1	9
Income	4.96	2.3	1	10
BMI	22.4	3.44	10.1	62.5
Self-rated health	74.35	18.69	0	100
Sleep quality	10.89	2.75	1	15
Chronotype	15.74	3.01	5	23
Depressive symptoms	5.87	5.53	0	27
Anxiety symptoms	4.3	4.52	0	21
Social support	15.45	3.81	3	21
Self-efficacy	10.89	2.43	3	15
Resilience	7.07	1.77	2	10
Family support	39.86	6.14	11	50
Adverse childhood experiences	0.44	0.87	0	4

4.2 Exploratory analysis

Figure 3 presents the correlation matrix of the main continuous variables. Perceived stress is most strongly and positively correlated with depressive symptoms and anxiety symptoms, indicating that higher levels of psychological distress are associated with higher stress. In contrast, perceived stress is negatively correlated with self-rated health, chronotype, self-efficacy, resilience, and family support, suggesting that better health and stronger psychosocial resources are associated with

lower stress. The matrix also shows moderate correlations among several psychosocial resource variables, as well as a strong correlation between depressive symptoms and anxiety symptoms. These preliminary patterns suggest that perceived stress is more closely related to psychological distress and psychosocial resources than to basic demographic characteristics alone.

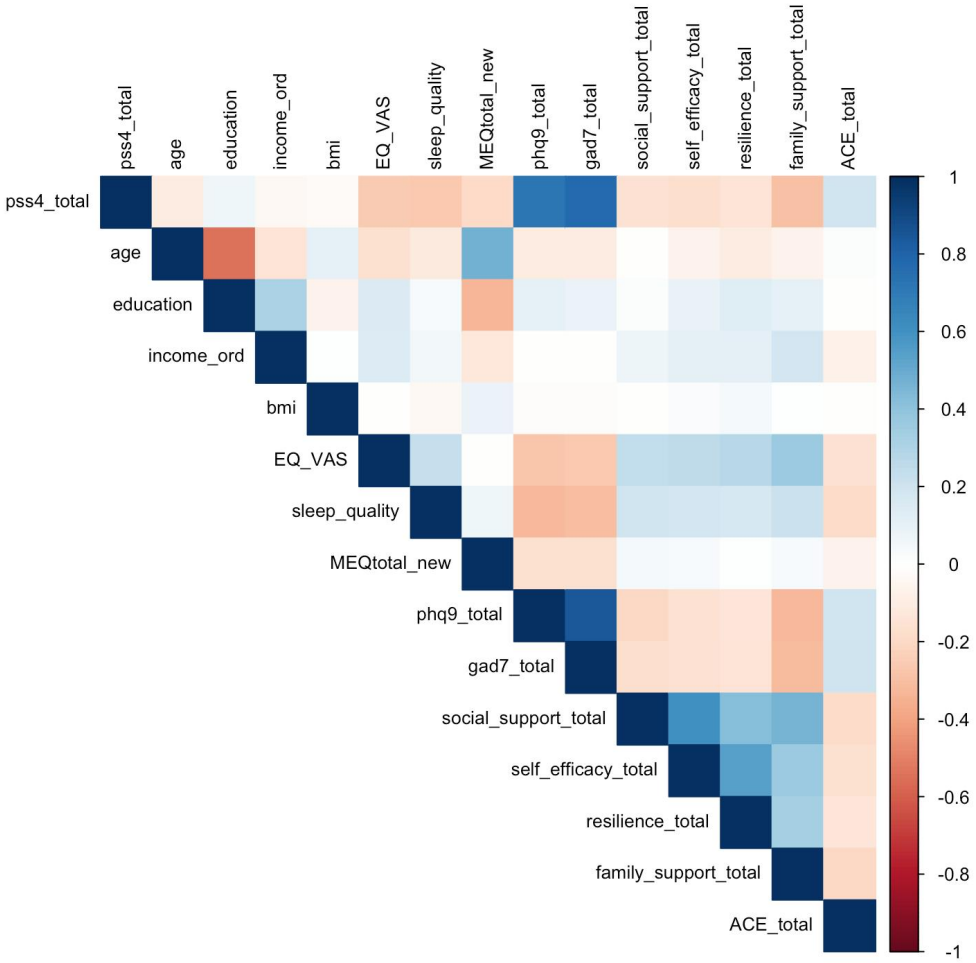

Figure 3: Correlation Matrix of Main Continuous Variables

4.3 OLS findings

Table 3 presents the OLS regression results for perceived stress using HC3 robust standard errors. The model explains a substantial proportion of the variation in perceived stress ($R^2 = 0.634$). Education, income, self-rated health, chronotype, self-efficacy, and family support are negatively associated with perceived stress, while

student status, flexible-job employment, depressive symptoms, anxiety symptoms, social support, major life events, debt, adverse childhood experiences, and drinking are positively associated with perceived stress. Among all predictors, anxiety symptoms and depressive symptoms show the strongest positive associations with perceived stress.

Table 4 reports the main diagnostic and robustness checks. The outcome variable is right-skewed, the residuals are not perfectly normal, and the Breusch–Pagan test indicates heteroskedasticity. Therefore, HC3 robust standard errors are used.

Table 3. OLS Regression Results for Perceived Stress (HC3 Robust Standard Errors)

Variable	Coefficient	Robust SE	t value	p-value	Sig.
Intercept	3.598	0.212	16.9579	< 0.001	***
Age	-0.001	0.001	-0.8034	0.421769	
Male	-0.006	0.029	-0.2068	0.836162	
Education	-0.049	0.009	-5.4268	< 0.001	***
Income	-0.025	0.006	-3.9477	< 0.001	***
Non-agricultural hukou	0.026	0.029	0.8982	0.36907	
Married	-0.112	0.101	-1.1040	0.269612	
Unmarried	-0.035	0.106	-0.3360	0.736858	
Widowed	-0.072	0.123	-0.5840	0.559213	
Student	0.194	0.052	3.7351	< 0.001	***
Unemployed	0.017	0.053	0.3255	0.744842	
Flexible job	0.084	0.041	2.0598	0.039422	*
Retired	-0.017	0.051	-0.3384	0.735048	
BMI	0.000	0.004	0.0918	0.926879	
Self-rated health	-0.004	0.001	-4.5930	< 0.001	***

Sleep quality	0.005	0.005	0.9060	0.364924	
Chronotype	-0.067	0.005	-12.9593	< 0.001	***
Depressive symptoms	0.143	0.006	24.0838	< 0.001	***
Anxiety symptoms	0.457	0.007	65.5771	< 0.001	***
Social support	0.055	0.005	11.2027	< 0.001	***
Self-efficacy	-0.097	0.008	-12.1094	< 0.001	***
Resilience	-0.003	0.009	-0.2838	0.776583	
Family support	-0.015	0.003	-5.4976	< 0.001	***
Lives alone	-0.051	0.039	-1.3014	0.193142	
Major life event	0.297	0.030	9.8586	< 0.001	***
Debt	0.078	0.029	2.7246	0.006442	**
Adverse childhood experiences	0.077	0.017	4.4735	< 0.001	***
Drinking	0.069	0.033	2.0607	0.039339	*
Smoking	-0.177	0.041	-4.3542	< 0.001	***
Chronic disease	0.008	0.033	0.2568	0.797352	

Model fit summary	
N	30054
Residual standard error	2.225
Multiple R-squared	0.634
Adjusted R-squared	0.6336
F-statistic	1793
Model p-value	< 0.001

Table 4. OLS Diagnostic and Robustness Checks

Check	Result
Outcome distribution	Perceived stress is right-skewed and lower scores are relatively concentrated.
Residual normality	Residuals deviate from normality, especially in the tails.
Breusch–Pagan test	BP = 752.19, df = 29, $p < 2.2e-16$.
Heteroskedasticity	Present.
Robust standard errors	HC3 robust standard errors were examined.
Robustness conclusion	The main conclusions remain largely similar after the HC3 adjustment.
Interpretation	OLS is retained as a baseline model, but results are interpreted with caution.

4.4 Lasso findings

Figure 4 shows the coefficients retained by the LASSO model at lambda.1se. Under this more parsimonious specification, anxiety symptoms, major life events, depressive symptoms, student status, and adverse childhood experiences remain positive predictors of perceived stress. By contrast, chronotype, self-efficacy, income, family support, and self-rated health remain negatively associated with perceived stress. Among the retained variables, anxiety symptoms have the largest positive coefficient, suggesting that they are the strongest predictor in the penalized model.

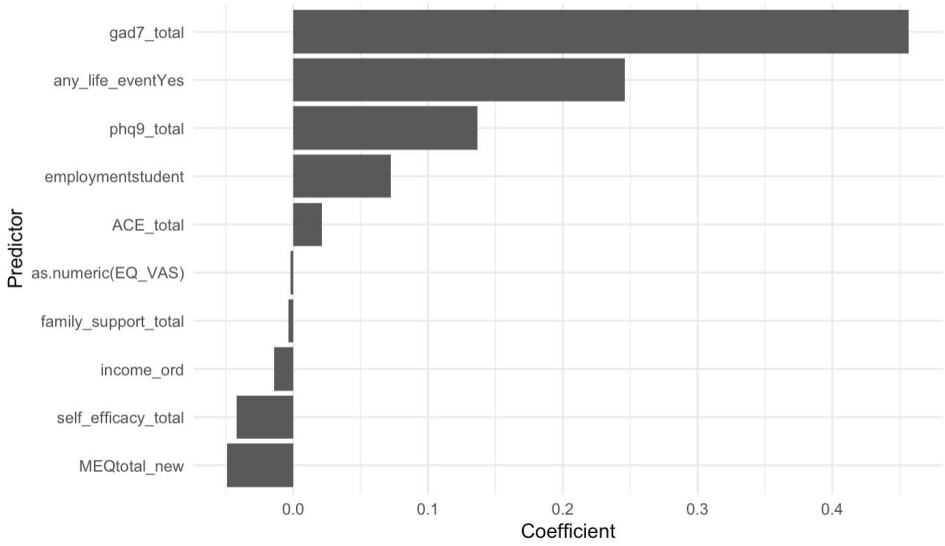

Figure 4: LASSO Coefficients at lamdba. 1se

4.5 Random Forest

I estimated two Random Forest models. The first model used the full set of predictors. The second model used only the predictors selected by LASSO. For both models, I split the data into a training set and a test set using an 80/20 split. I then used 5-fold cross-validation on the training set to tune the number of predictors randomly considered at each split and the minimum number of observations allowed in each terminal node. The number of trees was fixed at 500.

For the full Random Forest model, the best settings selected by cross-validation were 9 predictors considered at each split and 17 observations as the minimum terminal node size. On the test set, this model achieved a root mean squared error of 2.180, an R-squared of 0.651, and a mean absolute error of 1.640. For the LASSO-based Random Forest model, the best settings were 4 predictors considered at each split and 25 observations as the minimum terminal node size. On the test set, this model achieved a root mean squared error of 2.172, an R-squared of 0.654, and a mean absolute error of 1.653. The two models performed very similarly. This suggests that

Table 5. Random Forest Model Performance

Model	mtry	min_n	RMSE	R^2	MAE
Full Random Forest	9	17	2.18	0.651	1.64
LASSO-based Random Forest	4	25	2.172	0.654	1.653

the smaller set of predictors selected by LASSO still retained most of the predictive information for perceived stress.

4.6 SHAP findings

Table 5 reports the top 10 predictors ranked by mean absolute SHAP value in the full Random Forest model. Anxiety symptoms emerge as the most important predictor by a large margin, followed by depressive symptoms. Other important predictors include self-efficacy, social support, chronotype, family support, major life events, self-rated health, age, and resilience. Overall, the SHAP results suggest that perceived stress is driven more strongly by psychological distress and psychosocial resources than by basic demographic characteristics alone.

Table 6. Top 10 Predictors by Mean Absolute SHAP Value in the Full Random Forest Model

Rank	Predictor	Mean Absolute SHAP Value
1	Anxiety symptoms	1.445
2	Depressive symptoms	0.604
3	Self-efficacy	0.198
4	Social support	0.102
5	Chronotype	0.097
6	Family support	0.087
7	Major life event	0.082
8	Self-rated health	0.077
9	Age	0.072
10	Resilience	0.068

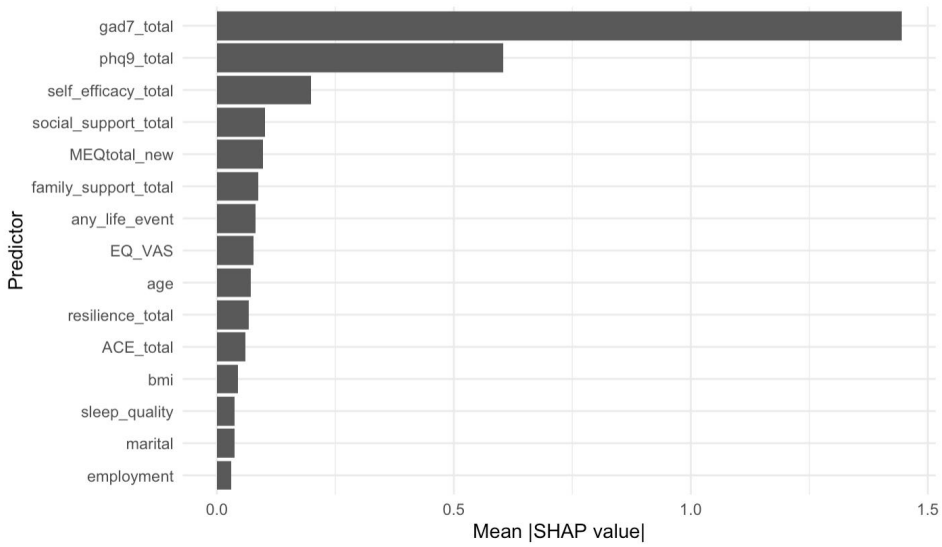

Figure 5: Top 15 Predictors by Mean Absolute SHAP Value

5. Discussion

This study examines which sociodemographic, psychosocial, and life-course factors matter most for perceived stress. Looking across the OLS, LASSO, and Random Forest results, a similar pattern keeps appearing. Basic demographic background does matter to some extent, but it does not seem to explain perceived stress as strongly as psychological distress, psychosocial resources, and difficult life experiences.

One of the clearest findings is the role of anxiety and depressive symptoms. These two variables stand out in all three models, and anxiety symptoms are especially strong. This suggests that perceived stress is closely tied to psychological distress[18]. But I do not think this should be read only as an individual emotional issue. It may also reflect the pressure and uncertainty people face in everyday life. In that sense, stress is not purely personal. It is also socially shaped.

Psychosocial resources also seem to matter a lot. In the OLS model, self-efficacy and family support are negatively associated with perceived stress. They also remain visible in the Random Forest and SHAP results. This pattern suggests that people who feel more capable of handling problems, or who have stronger support around them,

may be less vulnerable to stress. From a sociological perspective, this matters because coping resources are not evenly distributed. Some people simply have more support, more confidence, or more room to absorb pressure than others[19].

The life-course variables show a similar pattern. Major life events and adverse childhood experiences are both positively associated with perceived stress[20], and this does not disappear across models. That result is important because it suggests that stress is not only about current circumstances. Earlier hardship may continue to shape how people experience later difficulties[21]. So present stress may partly reflect accumulated disadvantage over time.

I also think the comparison across models is useful in itself. The LASSO-based Random Forest performs very similarly to the full Random Forest, which suggests that a smaller set of predictors can still retain most of the predictive information. At the same time, the SHAP results help make the full Random Forest more interpretable. Anxiety symptoms are clearly the most important predictor, followed by depressive symptoms. After that, variables such as self-efficacy, social support, chronotype, family support, major life events, self-rated health, age, and resilience also play a role. So although the methods are different, the overall story is fairly consistent.

There are also several limitations. First, the data are cross-sectional, so the results should not be interpreted causally. All of the main measures are also self-reported. That is especially relevant here because perceived stress, anxiety, and depression are conceptually close, so part of their association may reflect overlap in measurement. In addition, the models still do not include everything that may shape stress. Broader contextual factors, such as neighborhood conditions or workplace pressure, are not captured in the current analysis.

Even with these limitations, I think the study still points to something meaningful. Perceived stress does not look like a purely psychological outcome. It seems to be tied to emotional distress, unequal coping resources, and difficult experiences across the life course. Future research could examine this more clearly with longitudinal data and with better measures of social context.

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